

Network Level (II)

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Content

- **Network level**
 - **Routing activity**
 - **Preliminaries**
 - **Characterization**
 - **Routing**
 - **Routing protocols**
 - **RIP & OSPF**
 - **BGP & EGP**
 - **Network Congestion – general discussions**

Routing | Preliminaries

- The software part of the network level that chooses the path on which a received packet must be sent in order to reach the destination
- If datagrams are used, the routing decision must be taken for each packet
- If virtual circuits are used, the routing decision is taken when the circuit is established
- Requirements for a routing algorithm: correct, simple, robust, optimal, fast-converging
- Activities
 - Determine the optimal routing path
 - Packet switching

Terminology

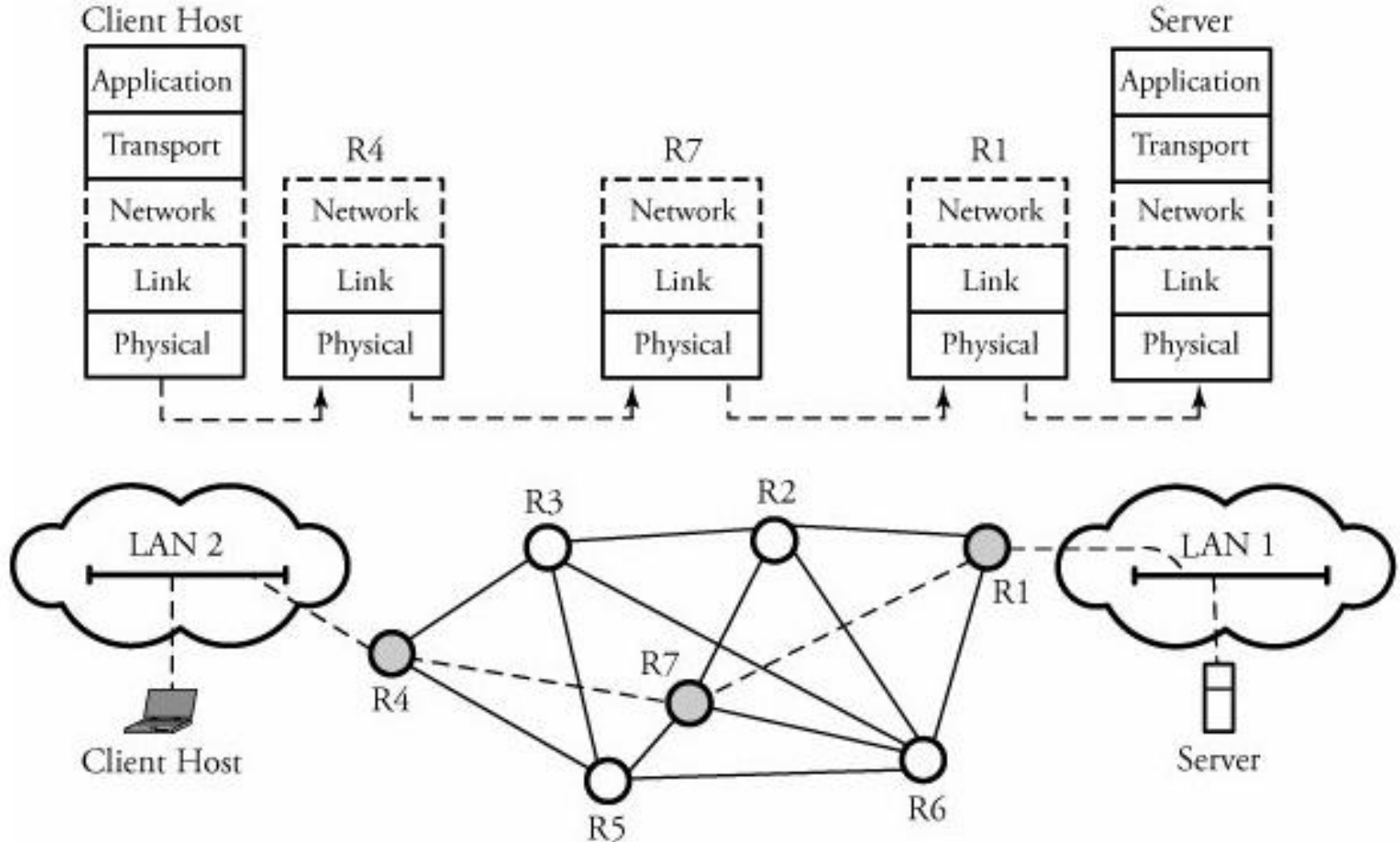
- **end systems** – network devices which do not have forwarding capabilities of packets to other sub-networks
- **intermediate systems** – network devices which have packet forwarding capabilities
 - Intradomain IS – communication within a routing domain
 - Interdomain IS – communication between routing domains
- **autonomous system** – a set of networks which share the same routing strategy and are managed by a unique administrative authority

Network level | ...let's remember

- At the network layer, the Internet can be seen as a collection of subnetworks or autonomous systems connected to each other
- IP makes this interconnection possible (see Course 2)
- The network level handles packet transmission from source to destination (mechanism which involves the passing through a series of intermediate nodes) => the network level is the lowest layer which deals with end-to-end transmission

Obs.: The role of host-to-network layer is to transport the frames from a point to another

Network level | ...let's remember

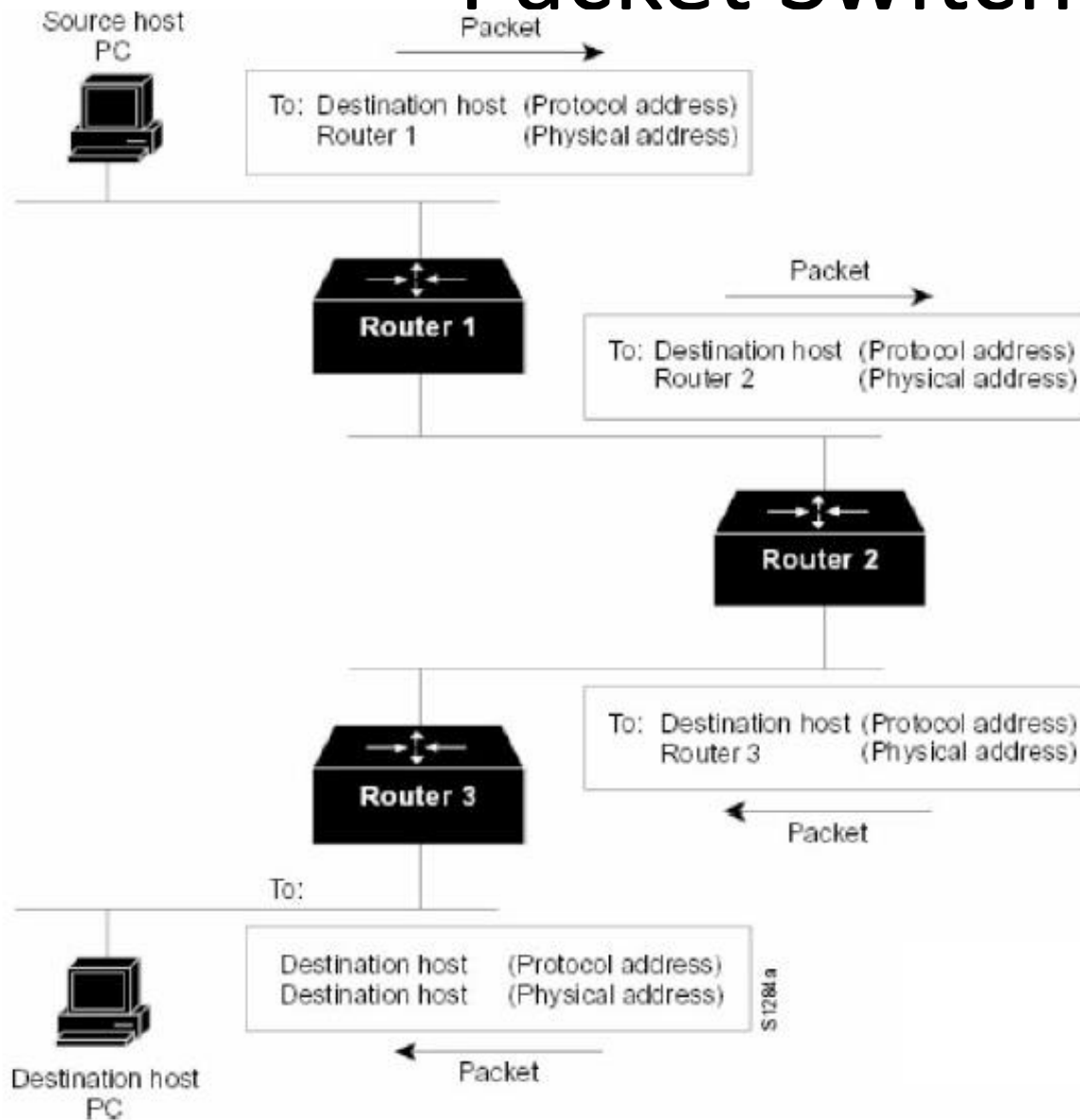


End-to-end communication between a client host and a server at network layer

Packet Switching

- A host needs to send a packet to another host
- The source host sends the packet to a router, using the router's MAC address; this packet contains the network address of the destination host
- The router examines the network address of the destination and if it does not know where to send the packet further, it will destroy it
- Otherwise, it will modify the address contained in the packet with the address of the next hop (intermediate transmission point – Intermediate System) and will send the packet to it
- If the next hop is not the final destination, then the process is repeated for another router etc.

Packet Switching

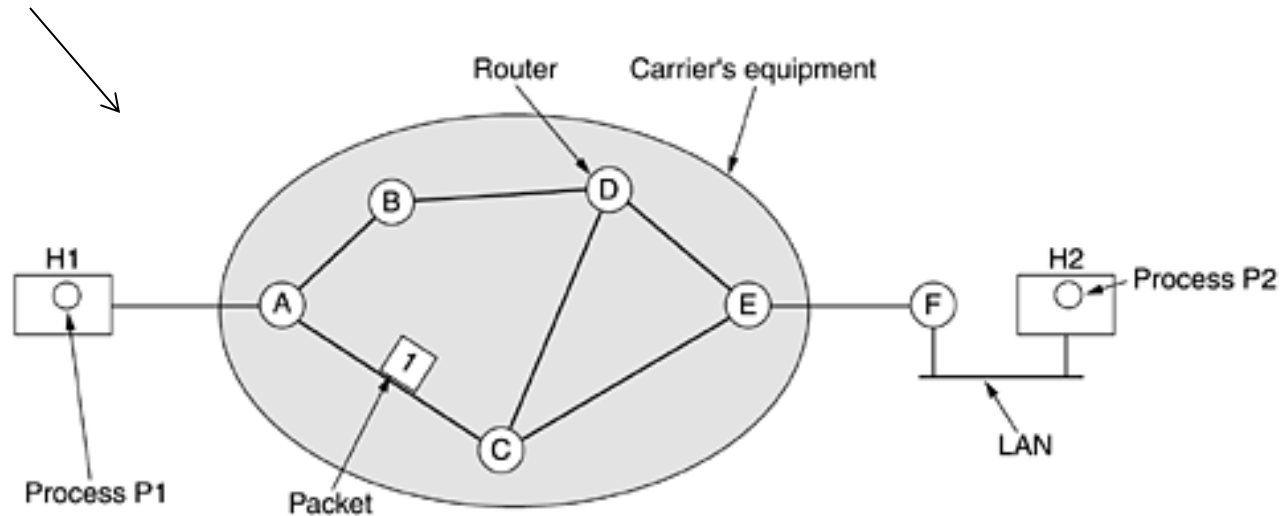


The switching
process

[Retele de calculatoare –
curs 2007-2008, Sabin Buraga]

Routing

The set of all communication subnets



Context for network layer protocols

Network layer:

- must know the routers' topology and choose the path through which a packet must be send to destination
- must make the choice in a way to avoid overloading communication lines or routers (details on next slides)

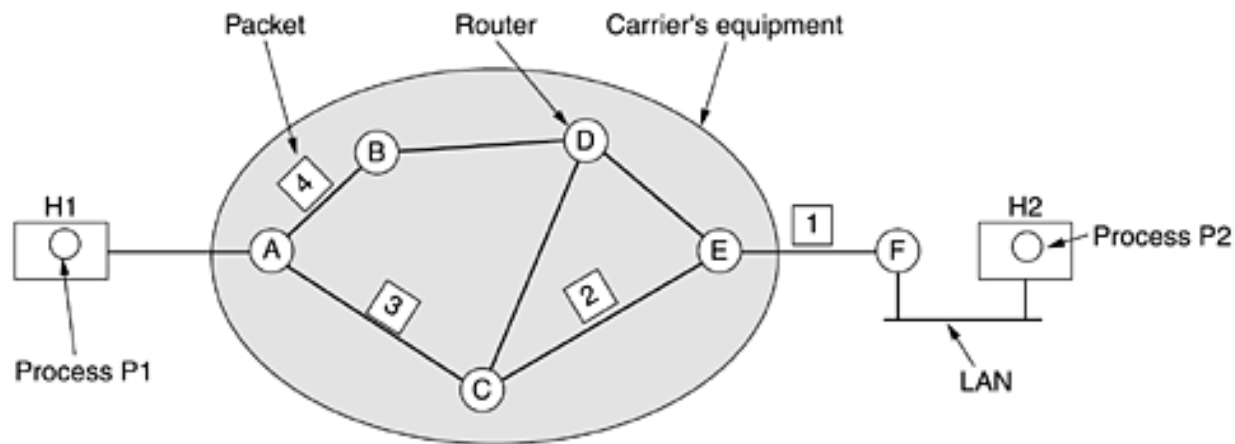


Determine the optimal routing path

[Computer Networks, 2021,
Andrew S. Tanenbaum et. al.]

Routing

- In case of connectionless services at network level, the packets (datagrams) are sent individually and are routed independently



The routing algorithms manage routing tables

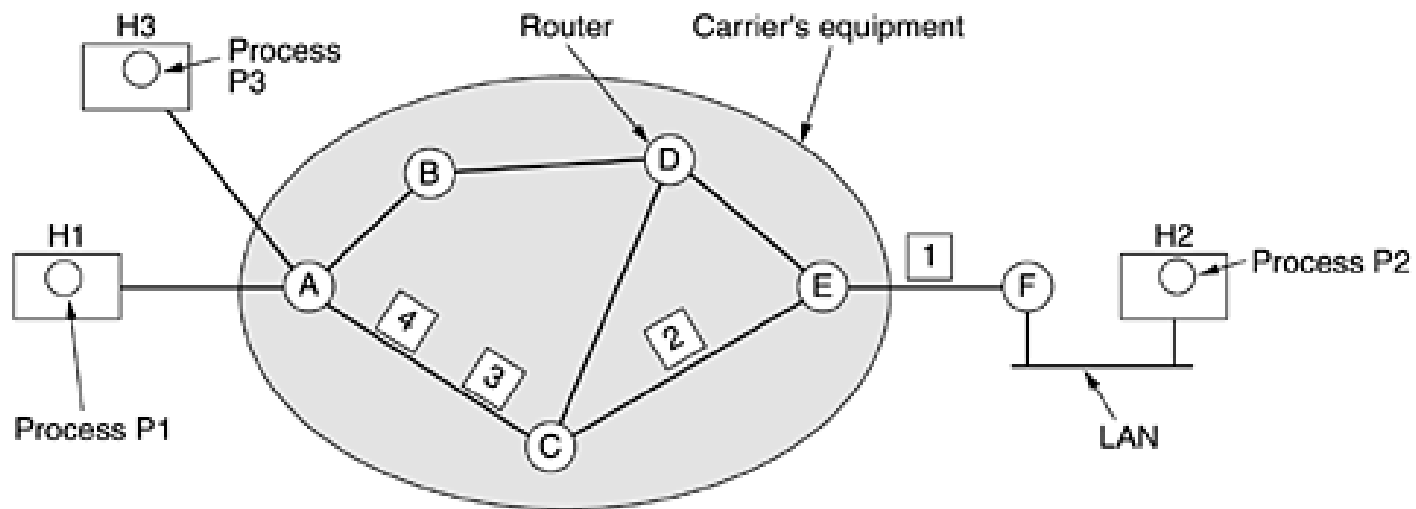
A's table				C's table		E's table	
initially	later						
A	-	A	-	A	A	A	C
B	B	B	B	B	A	B	D
C	C	C	C	C	-	C	C
D	B	D	B	D	D	D	D
E	C	E	B	E	E	E	-
F	C	F	B	F	E	F	F
Dest. Line							

Routing inside a datagram subnet

[Computer Networks, 2021,
Andrew S. Tanenbaum et. al.]

Routing

- In case of connection oriented services at network level, virtual circuits are used and the routing decision takes place when the circuit is established



Routing tables for routers A, C, and E:

A's table		C's table		E's table	
H1	1	A	1	C	1
H3	1	A	2	C	2
C	1	E	1	F	1
C	2	E	2	F	2

Label switching is indicated by arrows pointing to the 'In' and 'Out' columns of the tables.

Routing inside a
virtual-circuit subnet
(session routing)

[Computer Networks, 2021,
Andrew S. Tanenbaum et. al.]

Routing

Comparison between *datagram subnet* and *virtual-circuit subnet*

Issue	Datagram subnet	Virtual-circuit subnet
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

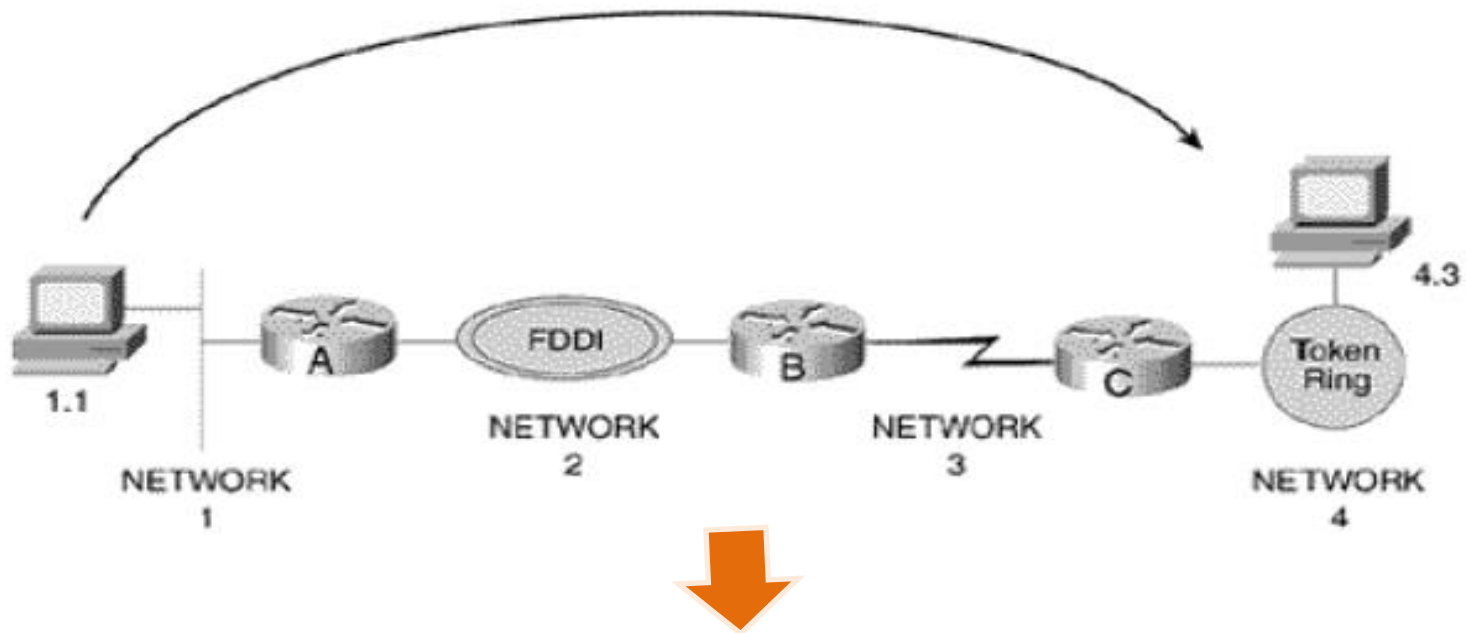
Routing

Determining the routing path

- For each path, a cost is determined (metric)
 - Path length, reliability, transmission delay, bandwidth, load, communication cost
- The routing algorithms initialize and keep (for each host) routing tables that contain routing information
 - Routes to specified hosts
 - Routes to specified networks
 - A default route

Routing

A router creates a logical path between subnetworks



An application which executes on host 1.1 does not need to know the path towards the application on host 4.3 in order to send data to it

[Retele de calculatoare –
curs 2007-2008, Sabin Buraga]

Routing

Characteristics of the routing algorithms:

- **Accuracy**: an algorithm must operate correctly and quickly to find the destination
- **Reduced complexity**: important for routers with limited hardware capabilities
- **Optimality**: the ability to find the optimal route
- **Robustness**: the capacity to correctly operate for a long period of time in different circumstances
- **Adaptability**: when a network error occurs, the algorithms must adapt (e.g., nodes failure or corruption of routing tables)
- **Convergence**: the algorithms must rapidly converge when routing messages announcing updates are distributed
- **Load balancing**: an algorithm weights different routing possibilities in order to avoid slow links or congestions

Routing

- Abstract notions
 - Network = graph
 - Routing = finding the shortest path from a source node to a destination node

Routing types:

- **Centralized** – the shortest path can be determined by having all the information about the network -> **link state routing algorithms**
- **Decentralized** – the shortest path is determined iteratively, distributed (there is not a single node which possesses complete information about the costs of the network links) -> **distance vector routing algorithms**

Routing

- Routing using **link state**
 - The network's topology & the costs for all links are known
 - A node must know the identities & costs of its neighbor nodes
 - Each node broadcasts the identities and costs of all links from it to others

Routing

- **Distance vector** routing
 - Each node receives information from its neighbor nodes, does some calculations and distributes the results back to its direct neighbors – the algorithm is distributed and asynchronous
 - Each node maintains a **distance table**
 - **X**: the node that wants to do a routing to node **Y** via neighbor node **Z**
 - **$D^X(Y,Z)$** : the sum of the direct link's cost between **X** and **Z** (**$c(X,Z)$**), plus the current cost of the shortest path from the neighbors of **Z** to **Y**:
$$D^X(Y,Z) = c(X,Z) + \min_w \{D^Z(Y,w)\}$$
 - A node's routing table can be built by knowing its distance table

Routing

Routing algorithms – classification:

- **Static (nonadaptive)**
 - The topology of the links is loaded for a period of time in the routing tables of each node
 - Disadvantages:
 - The network must be of reasonable (optimal) size in order to be controllable
 - If network errors occur, there can not be an immediate reaction
- **Dynamic (adaptive)**
 - The network state is “learned” through the communication of the routers with their neighbors; the state of each network region is propagated in the whole network after all the nodes update their routing tables => each router can find the best path based on the information from the neighbor nodes

Routing

Routing algorithms – classification:

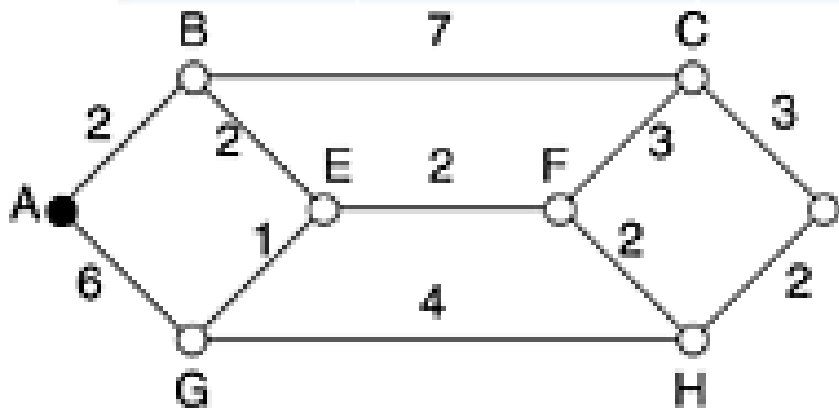
- Static (nonadaptive)
 - Shortest path routing
 - Flooding
 - Deflecting routing (or hot-potato routing)
- Dynamic (adaptive)
 - Distance vectors
 - Using link state
 - Hierarchical routing
 - Through broadcast
 - Through multicast

Routing

Routing algorithms – classification:

- **Static** (nonadaptive)
 - **Shortest path routing**
 - Dijkstra's algorithm (calculates the minimum cost path)
 - Is used by OSPF protocol

“shortest path”: no. of hops => ABC and ABE are equal

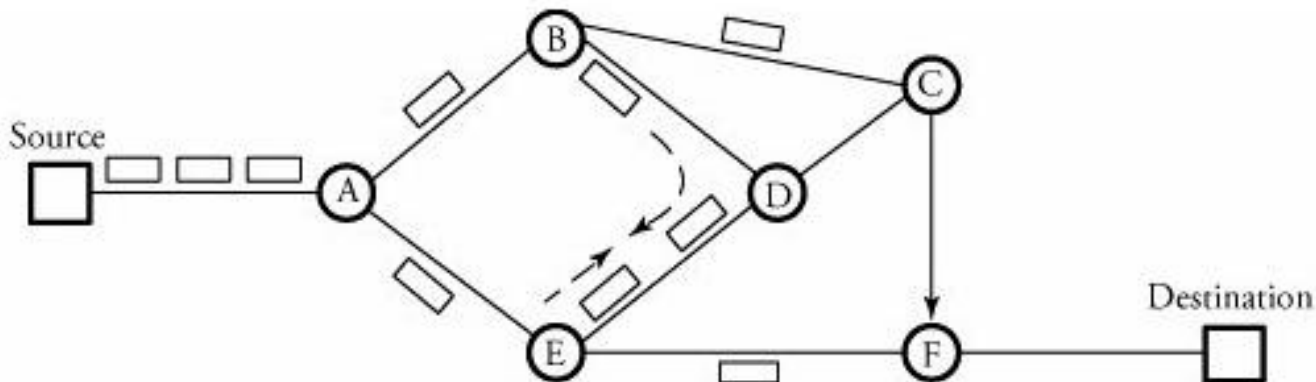


Other possible metrics: geographic distance, bandwidth, communication costs etc.

Routing

Routing algorithms – classification:

- Static (nonadaptive)
 - **Flooding**
 - A received packet is copied and sent through all communication links (except the one it came from)
 - The problem: packet reflection (a node can receive an unwanted copy of a packet)
 - Usages of flooding algorithms: military applications, distributed databases etc.



Routing

Routing algorithms – classification:

- Static (nonadaptive)
 - **Deflection routing**
 - At each step, a packet is examined in relation to the destination address; if the requested link is free, the packet is sent, otherwise it is deflected to another communication link chosen randomly
 - A packet has a field associated, containing a value representing a priority, which can help it in the future to win the “dispute” with other packets

Routing

Routing algorithms – classification:

- **Dynamic** (nonadaptive)
 - **Distance vector**
 - Each router maintains a table (vector) with the distance and the communication link towards the destination; the tables are updated with information received from the neighbors

Bellman-Ford algorithm

- Used by RIP, BGP, IGRP protocols

Example:

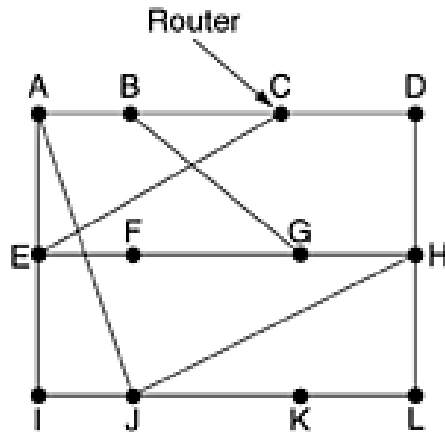
- As a metric we consider the delay (msec);
- The routers will know the delays associated with their neighbors

Routing

Routing algorithms – classification:

- Dynamic (adaptive)
 - Distance vector

Example:



J wishes to compute the route to G

J -> A -> G = 26 (18 + 8) msec



J->H->G 18 msec

To	A	I	H	K	New estimated delay from J ↓ Line	
A	0	24	20	21	8	A
B	12	36	31	28	20	A
C	25	18	19	36	28	I
D	40	27	8	24	20	H
E	14	7	30	22	17	I
F	23	20	19	40	30	I
G	18	31	6	31	18	H
H	17	20	0	19	12	H
I	21	0	14	22	10	I
J	9	11	7	10	0	–
K	24	22	22	0	6	K
L	29	33	9	9	15	K

JA delay is 8	JI delay is 10	JH delay is 12	JK delay is 6
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Vectors received from J's four neighbors

New routing table for J

[Computer Networks, 2021,
Andrew S. Tanenbaum et. al.]

Routing

Problem: in case of the distance vector algorithm, on each update of the routes, the routing tables must be sent to each neighbor; some packets containing information regarding routing go through the same route they came from (reverse route)

Question: Can reverse routes be avoided?

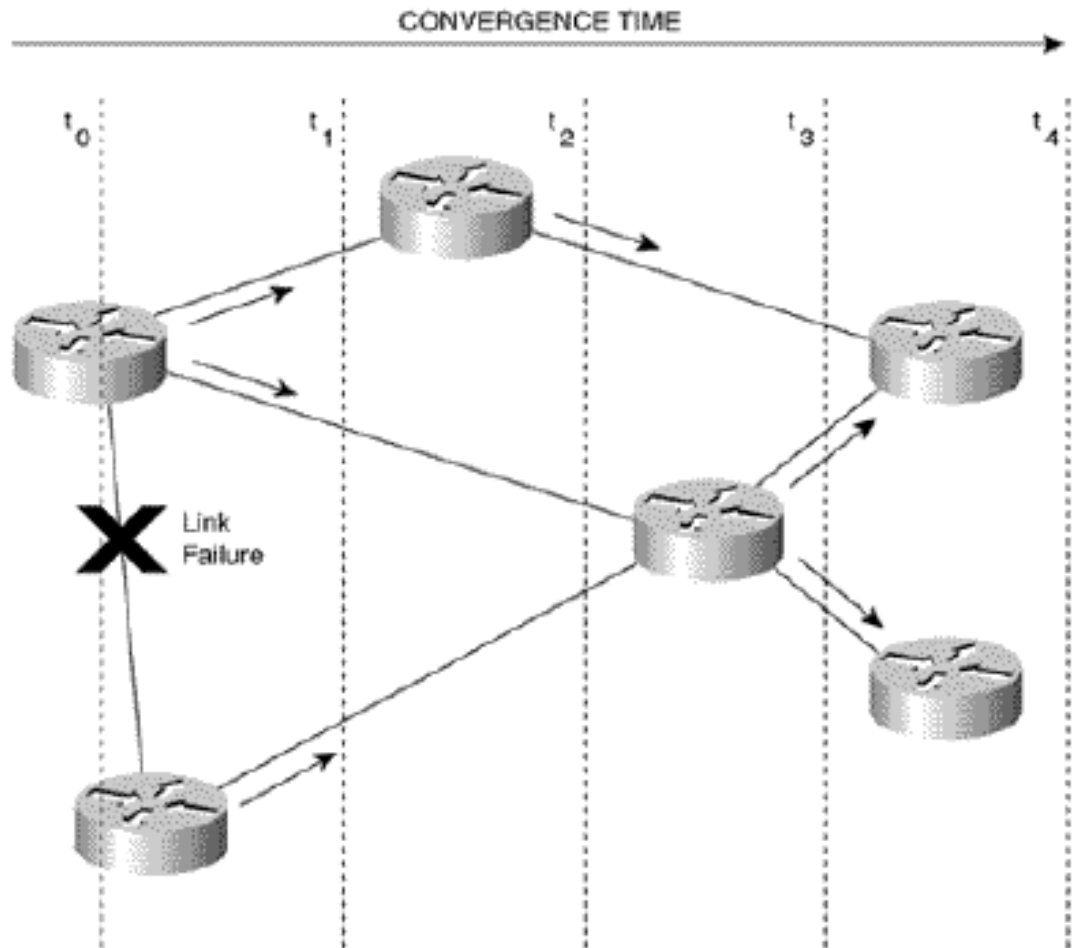
Answer: using the ***split horizon*** technique

- When the router sends route updates using a certain network interface, they will not be sent to the networks whose routes were learned from updates received via that interface

Routing

Problem:

Topology
modification
(routing
algorithms
convergence
depreciation)



[Retele de calculatoare –
curs 2007-2008, Sabin Buraga]

Routing

Routing algorithms - classification:

- Dynamic (adaptive)
 - **Using link state**

Each router must:

- Discover its neighbors and “learn” their network addresses
- Measure the delay or the cost associated with each neighbor
- Create a packet through which it announces everyone about what it learned

Dilemma: when must the packets be created? (e.g., periodically or when a special event occurs)

- Send the packet
- Calculate the shortest path to each router

Routing

Routing algorithms - classification:

- Dynamic (adaptive)

- **Hierarchical routing**

Necessity: in large networks it is not feasible for a router to have an entry about every other router;

Mechanism: Routers know details about a region, but they do not possess information about the internal structure of other regions

Obs.:

- For large networks, a level 2 hierarchy is not sufficient, thus the regions are grouped in clusters, the clusters in zone, the zones in groups etc.
 - Which is the optimum number of levels?

For a subnet with N routers, the optimum number: **$\ln N$**
[Kamoun&Kleinrock, 1979]

Routing

Routing algorithms - classification:

- Dynamic (adaptive)

- ***Broadcast routing***

Use: stock updates (on stock exchanges), multimedia streaming, weather reporting distribution service etc.

Ways:

- **The source sends a distinct packet to each recipient**

Obs.: Ineffective method: does not use the bandwidth; the source must know the addresses of all the recipients

- **Flooding** – useful when other methods can not be applied

Problem: many packets are generated and a lot of bandwidth is used

Routing

Routing algorithms - classification:

- Dynamic (adaptive)
 - ***Multicast routing***

Example:

A process wants to send a message to a group of processes which implement a distributed database system

Obs.: Broadcast can be used, but sometimes the information is not intended to be seen by everyone

Mechanism: the router will periodically interrogate the hosts that belong to a group; then the information is propagated to the routers

Routing

Example:

```
[adria@thor ~] $ netstat -r
Kernel IP routing table
Destination      Gateway          Genmask          Flags      MSS  Window  irtt  Iface
webapps.vpn      *               255.255.255.255  UH          0  0        0  tun1
172.18.100.2     *               255.255.255.255  UH          0  0        0  tun0
85.122.23.0      *               255.255.255.128  U           0  0        0  eth2
172.18.100.0     172.18.100.2    255.255.255.0    UG          0  0        0  tun0
172.17.17.0      webapps.vpn     255.255.255.0    UG          0  0        0  tun1
default          gw-info-c.uaic. 0.0.0.0          UG          0  0        0  eth2
[adria@thor ~] $ netstat -rn
Kernel IP routing table
Destination      Gateway          Genmask          Flags      MSS  Window  irtt  Iface
172.17.17.1      0.0.0.0          255.255.255.255  UH          0  0        0  tun1
172.18.100.2     0.0.0.0          255.255.255.255  UH          0  0        0  tun0
85.122.23.0      0.0.0.0          255.255.255.128  U           0  0        0  eth2
172.18.100.0     172.18.100.2    255.255.255.0    UG          0  0        0  tun0
172.17.17.0      172.17.17.1     255.255.255.0    UG          0  0        0  tun1
0.0.0.0          85.122.23.126   0.0.0.0          UG          0  0        0  eth2
```

[https://docstore.mik.ua/oreilly/networking_2ndEd/tcp/ch02_04.htm]

Example: For the destination 172.17.17.0 the used router (gateway) is 172.17.17.1 ; Gateway = 0.0.0.0 -> local network interface

Flags: U(*up*) – the route is operational; H – indicates a route to a certain host; G – the route uses an exterior gateway

Routing tables creation

- Static routes: **route** UNIX command

- Discover a router through ICMP

- Broadcast protocol which discovers the routers of a local network

Routing

Routing protocols – classification:

- **Intradomain routing protocol** – routes packets inside a domain
 - RIP (Routing Information Protocol)
 - OSPF (Open Shortest Path First)
- **Interdomain routing protocol** – routes packets between domains
 - BGP (Border Gateway Protocol)
 - EGP (Exterior Gateway Protocol)
 - RFC 827, 904
 - it is not used anymore, being replaced by BGP

Routing

RIP (Routing Information Protocol)

- RFC 1058, 2453
- Functioning mechanism:
 - Bellman-Ford algorithm is used (for hosts and routers)
 - For each router, a vector is created, which contains the cost of the route and other information
 - If changes occur in a point, these are periodically propagated to the neighbor routers and hosts of that point

Routing

RIP (Routing Information Protocol)

- Uses IP messages
- Each router sends a broadcast containing the entire routing table of the router – on each 30s
- An entry in a RIP routing table contains:

- IP address
- Metric (number of hops: 1-15)
- Timeout (in seconds)

Routing table A: node B is 1 hop distance (direct connection), node C is 2 hops

- Networks directly connected have metric = 1 (one hop)



- If a route timeouts, the metric becomes 16 (no connection) and the route is deleted after 1 minute

Routing

RIP (Routing Information Protocol)

- If a routing information is changed (e.g., a link or a router fails), the propagation of this change takes place slowly – RIP has slow convergence
- RIP:
 - Is a mature protocol, stable, widely supported and easy to implement
 - Is recommended to be used by autonomous systems with reduced size and without redundant routes
 - In practice it is replaced by OSPF in many situations

Routing

OSPF (Open Shortest Path First)

- RFC 2328
- Each router that uses OSPF knows the state of the entire network topology (algorithm that uses link state) and sends updates to all routers



- This generates additional traffic, which can lead to congestions
OSPF allows the traffic to be distributed on routes with similar costs (load balancing)

OSPF supports type-of-service (ToS) routing

- the IP protocol contains the field ToS (generally not used)
- Faster convergence
- Supports the use of many types of metrics

Routing

OSPF (Open Shortest Path First)

- Operates in a hierarchy of network entities:

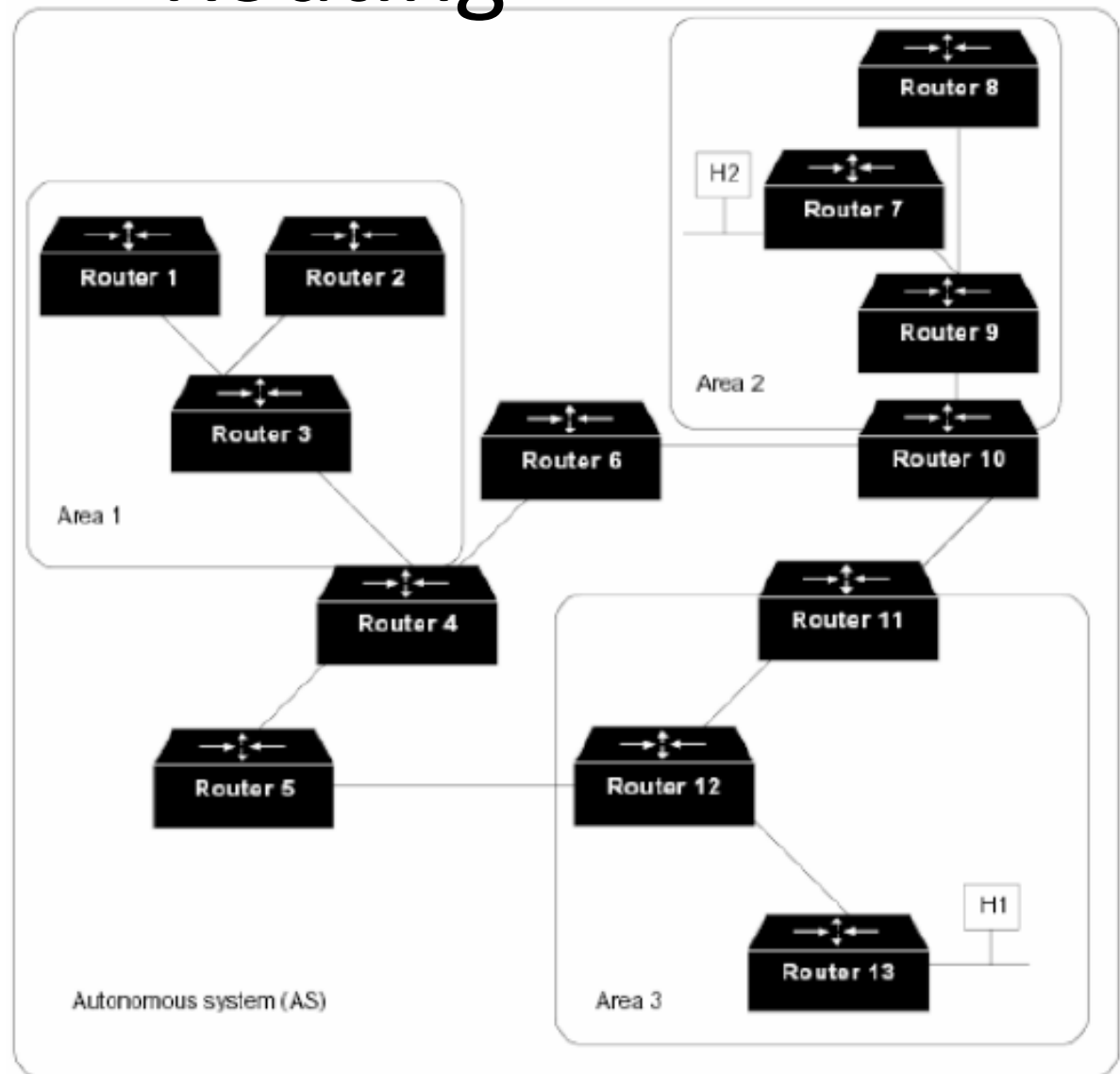
Motivation: large networks => a router cannot know the whole topology

- **Autonomous system** (AS) – collection of networks which share the same routing strategy
- An AS is divided in **areas** – contiguous groups of networks and hosts; the routers have the same information regarding the topology and use the same algorithm
- **Backbone** (or **area 0**) – responsible with the distribution of routing information between domains; any router connected to two or more domains is part of the backbone (these routers will use algorithms corresponding to each domain)

Routing

OSPF (Open
Shortest
Path First)

An AS and its
domains,
connected
through
routers



Routing

OSPF (Open Shortest Path First)

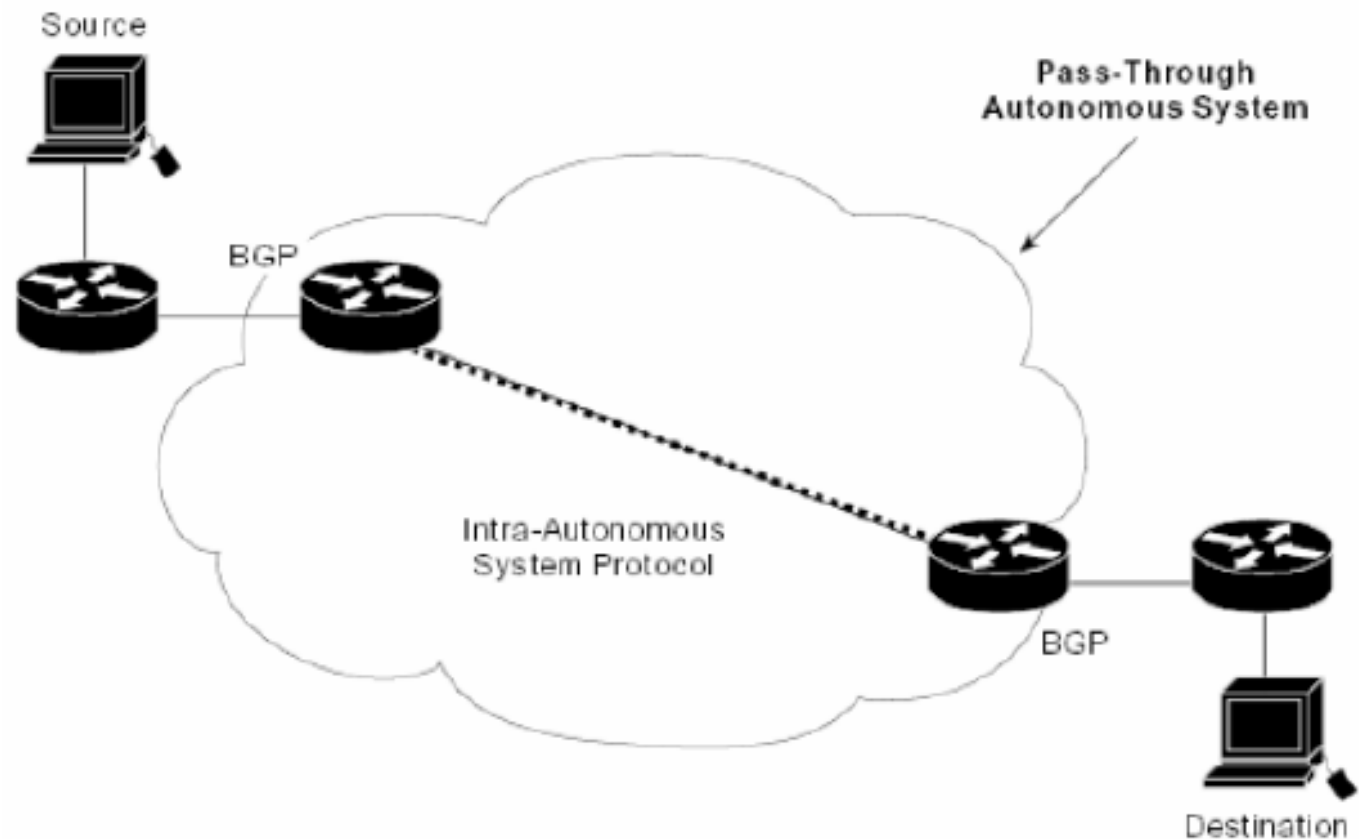
OSPF message types:

Message type	Description
Hello	Used to discover who the neighbors are
Link state update	Provides the sender's costs to its neighbors
Link state ack	Acknowledges link state update
Database description	Announces which updates the sender has
Link state request	Requests information from the partner

- Using a “hello” message, a routers discovers its neighbors (e.g., all LAN routers)
- Each router periodically floods with a message (that has associated a sequence number) of type *Link state update*; for these messages, *Link state ack* confirmations are sent
- *Database description* provides the sequence numbers associated to the link entries that the transmitter have

Routing

BGP (Border Gateway Protocol)



[Retele de calculatoare –
curs 2007-2008, Sabin Buraga]

Routing

BGP (Border Gateway Protocol)

- Used for the communication between routers in different autonomous systems
- Major functions:
 - *Neighbor relationship* – refers to the agreement between the routers belonging to two autonomous systems to exchange information based on some rules (a router can refuse establishing such a relationship depending on: domain's rules, overload etc.)
 - *Neighbor maintenance* - the routers will send keep-alive messages
 - *Network maintenance* – each router keeps a database with the existing subnets for an efficient routing in that subnet

Routing

BGP (Border Gateway Protocol)

- There are four types of BGP packets:
 - *Open*: used for establishing a relationship between two routers
 - *Update*: contains updates about routes
 - *Keep-alive*: used for confirming previously established relationships
 - *Notification*: used when errors occur
- BGP router pairs communicate between them using TCP connections
- BGP is a protocol based on distance vector, with the following differences:
 - beside the cost associated with a destination, the path to it is also stored
 - the neighbors are not provided only with the estimated cost, but also with the exact path
- RFC 1771-1774, 4271

Routing

Other protocols:

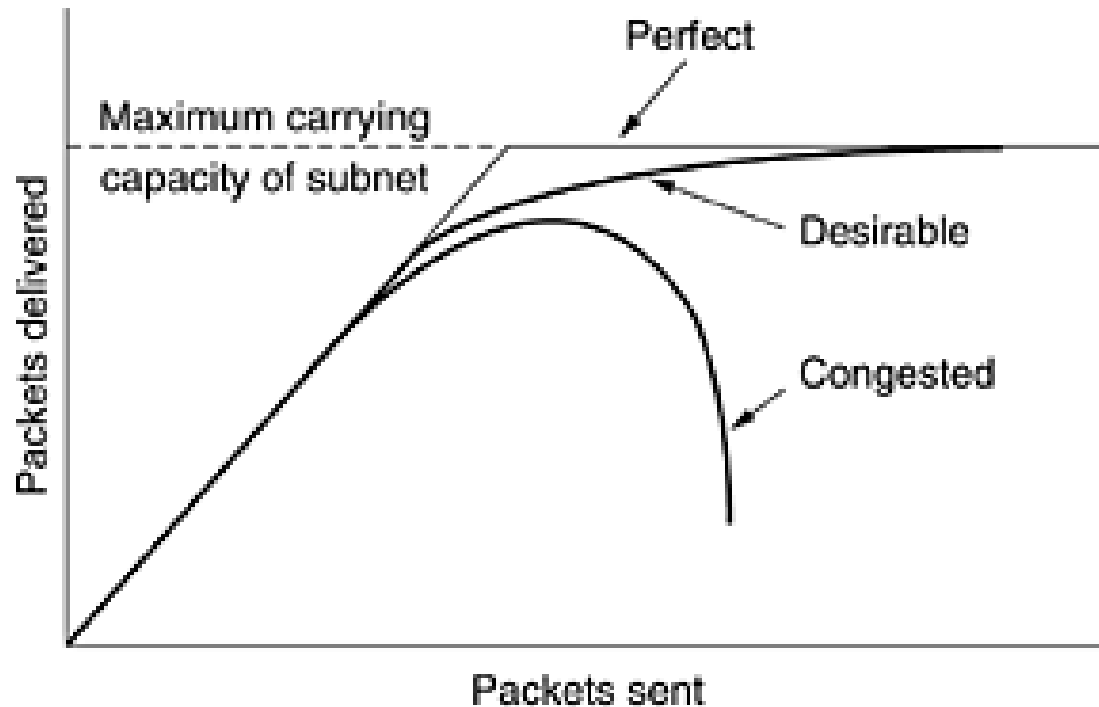
- Interior Gateway Routing Protocol (IGRP)
 - a RIP improvement by CISCO
 - Enhanced IGRP (EIGRP)
 - Simple Multicast Routing Protocol (SMRP)
 - Multimedia streams routing from Apple (via AppleTalk)
- Obs.: Since 2009, AppleTalk is not supported anymore, being replaced by TCP/IP
- Resource Reservation Protocol (RSVP) (RFC 2205)
 - It is not a routing protocol, but offers similar functionalities
 - Ensures IP QoS (quality of service)

Routing | overview

- Internal routing:
 - RIP (Routing Information Protocol)
 - IGRP (Interior Gateway Routing Protocol)
 - EIGRP (Enhanced IGRP)
 - OSPF (Open Shortest Path First)
 - IS – IS (Intermediate System to Intermediate System) for ISO/OSI
- External routing:
 - BGP (Border Gateway Protocol)
 - EGP (Exterior Gateway protocol)

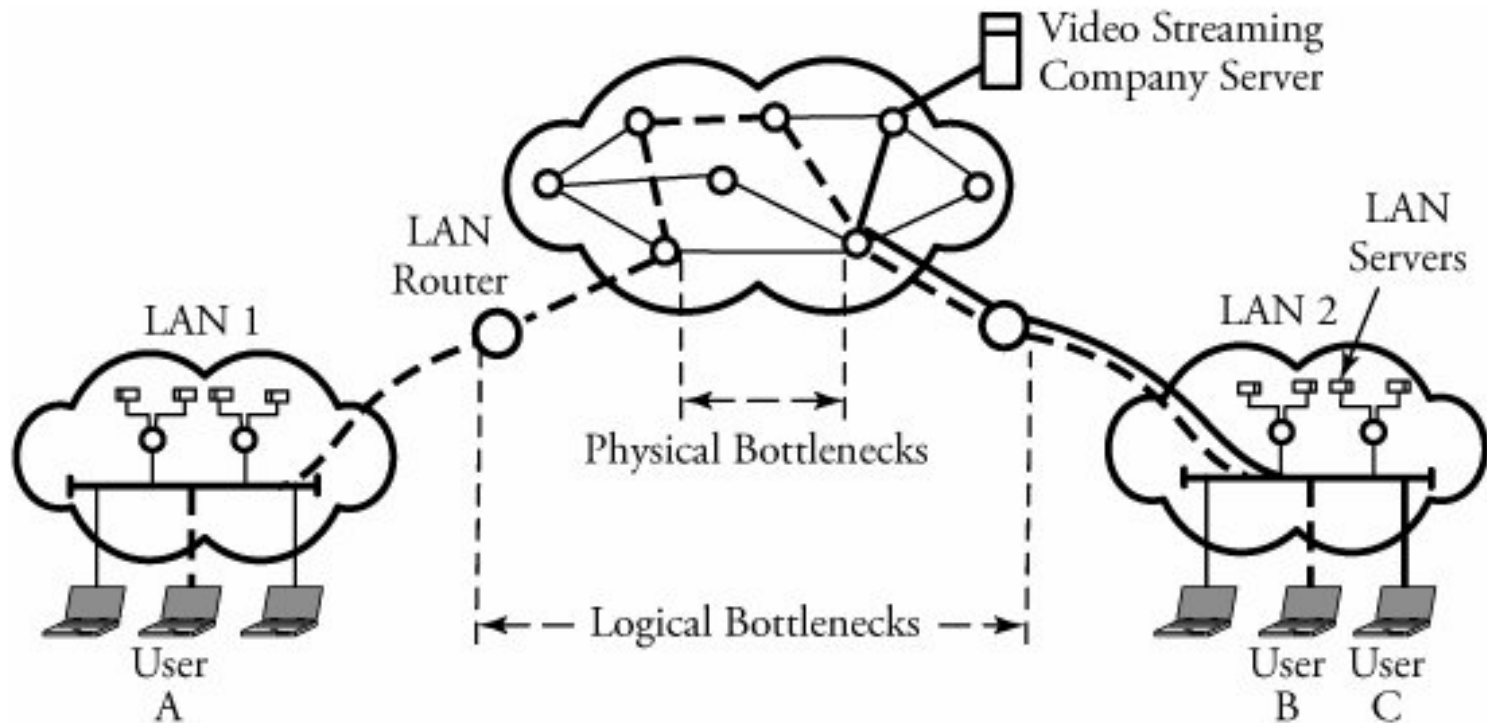
Congestion | Discussions

- Occurs when a network's resources are overloaded



In case of high traffic, congestion can appear and the performance drops sharply

Congestion | Discussions



Congestion can appear:

- At data link layer: when there is not enough bandwidth
- At network layer: when the queue from the nodes can not be controlled
- At transport layer: when the logical link between two routers having a communication session can not be controlled

Congestion | Discussions

- Congestion control – solutions
 - **Open-loop**: the solution means preventing the occurrence of congestions through good decisions and design
 - **Close-loop**
 - System monitoring for detecting congestions

Metrics: the percent of removed packets due to lack of space in *buffer*, packet delay etc.
 - Sending this information to the nodes that can make decisions
 - Adjust the operations to correct the problem

Congestion | Discussions

Obs.:

congestion control != stream control

- Congestion control ensures that the network has the capacity to transport the traffic; involves the actions of all hosts and routers
- Stream control handles the point-to-point communication between a sender and a receiver and ensures the fact that the sender does not transmit data faster than the receiver can process

Summary

- **Network level**
 - **Routing activity**
 - **Preliminaries**
 - **Characterization**
 - **Routing**
 - **Routing protocols**
 - **RIP & OSPF**
 - **BGP & EGP**
 - **Network Congestion – general discussions**

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Questions?

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